

SOLVING INVENTORY INEFFICIENCIES USING SQL

Executive Summary Report

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Urban Retail Co. is facing a serious inventory management problem across all 20 stores in 4 regions. Current performance is far below industry standards, and urgent action from company leaders is needed to stop further financial losses.

Key Crisis Indicators

- **Inventory Turnover:** 0.61-0.71 (Industry Standard: 4-12)
- **Forecast Accuracy:** 2-40% (62-98% error rates)
- **Working Capital Locked:** 500K+ in slow-moving inventory
- **Stockout/Overstock Pattern:** 85% of stores critically imbalanced

Common Critical Products across all stores

- **Understocked:** P0016, P0046, P0057, P0061, P0125, P0129, P0166, P0175, P0187
- **Overstocked:** P0031, P0068, P0085, P0096, P0116, P0153, P0183

Financial Impact Analysis

Category	Price gap	Revenue loss (annual)	Recommendation
Clothing	-4.31 (7.8%)	85K+	Increase prices 6-8%
Groceries	-4.23 (7.6%)	45K+	Match competitor pricing
Electronics	-4.10 (7.5%)	95K+	Dynamic pricing implementation
Furniture	-4.16 (7.5%)	35K+	Premium positioning strategy
Toys	-4.15 (7.5%)	25K+	Seasonal price optimization

Total revenue recovery potential : 285K+ annually

Portfolio misallocation

Category	Current %	Optimal %	Action required
Electronics	39% (952 SKUs)	25%	Eliminate 300 SKUs
Clothing	32% (779 SKUs)	25%	Reduce by 200 SKUs
Groceries	7% (160 SKUs)	25%	Expand by 200 SKUs
Furniture	20% (587 SKUs)	20%	Maintain current
Toys	2% (206 SKUs)	5%	Optimize seasonally

Inventory Analysis

There is no consistency in the units ordered from suppliers. This indicates inefficient inventory management, which increases the risk of understocking/overstocking.

In many cases, there is not much difference between units sold and units ordered, indicating the dependency on number of units sold to place order for restocking.

To avoid this, keep records of daily inventory, and the units sold for each product. Always keep a safety buffer of units based on demand to avoid understocking.

A significant forecasting error has been identified. So there is a need to fix the system.

Recommendations / Action Plan

Action	Impact
Increase prices by 4-6%	Extra income
Give 15-20% discount on overstocked items	Overstock clearance, amount recovery
Order urgently needed products	Avoid running out of top-selling items
Move extra stock between stores	Balance inventory
Fix inventory forecasting system	Make stock planning more accurate

Use smart reorder levels	Restock only when needed based on real sales
Remove slow-moving products	Save huge amount of money blocked in storage costs
Use real-time inventory tracking	Always know stock levels at all stores